

The Audit — System Architecture

Audit of an intern's tokenizer-fertility & serving-capacity report (REPORT_v0) before leadership routing/capacity decisions · repo: `github.com/Manureddy148/AI_Task2` · audit 2026-08-31, docs compiled 2026-09-04

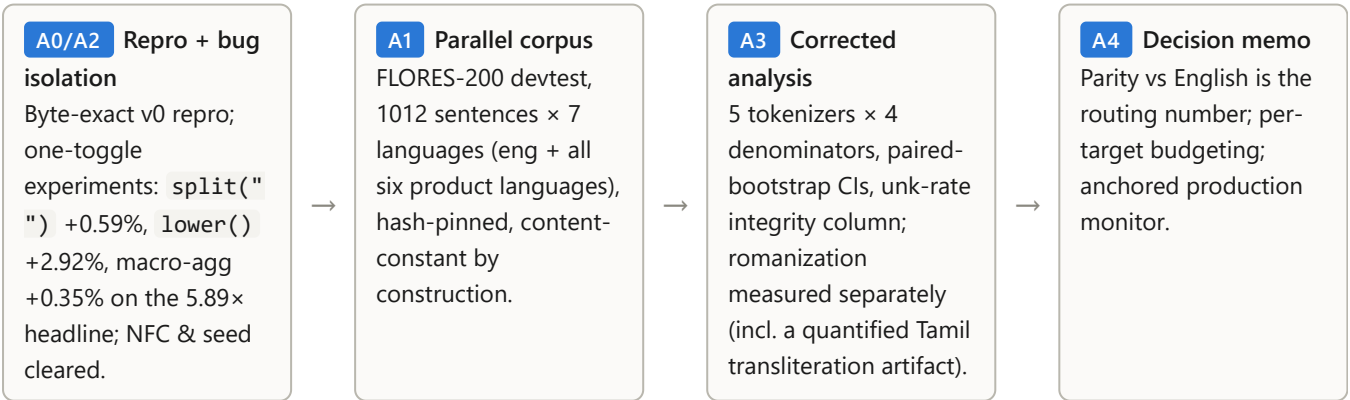
1 What is audited, and what the audit ships

REPORT_v0 makes two families of claims: **§1** "Hindi costs ~6× English per token-word; the two metrics confirm each other; it's a property of the script — route Indic traffic and budget 6×", and **§2** "longer prompts give better throughput; plan ~1600 tok/s per L4, linear to ~3200 at batch 48". Every claim rests on four shipped artifacts (`fertility.py`, `corpus_sample/`, `bench/model_spec.md`, `bench/bench_log.csv`), so every claim is checkable — and gets checked. The audit's product is **REPORT_v1** (the corrected report leadership can actually use) sitting on a fully machine-verified evidence chain.

2 Evidence discipline — the five design rules

Rule	Mechanism
Pin the baseline first	REPORT_v0's numbers reproduced byte-exact from the shipped script + corpora before any claim; all deltas are then attributable.
One variable at a time	Each suspected bug is a single toggle against the as-shipped baseline (<code>experiments.py</code>); no flaw claimed without a measured Δ or a demonstrated conceptual failure on data.
Cleared items are deliverables	Suspicious-but-fine items (<code>random.seed</code> , NFC) are cleared by experiment — hash-identical output, 0.00% deltas — not flagged.
Scripts over prose	Every table is emitted by a version-controlled script; live-defense "change this input" is served by CLI flags (<code>--corpus LANG=PATH</code> , <code>--hin</code> , <code>--log</code> , <code>STARTER_KIT</code> env).
Numbers under test	<code>verify.py</code> re-derives and asserts every published figure — 50 checks — run locally and in CI on every push. A prose number with no assertion is treated as unpublished.

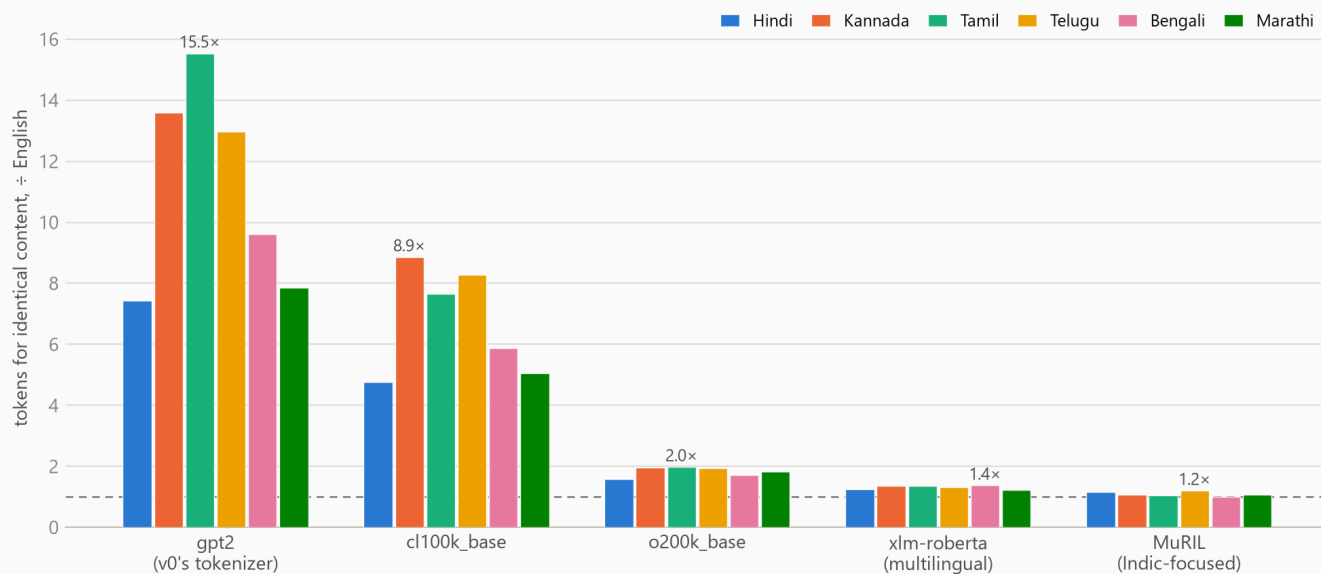
3 Part A — tokenizer audit pipeline



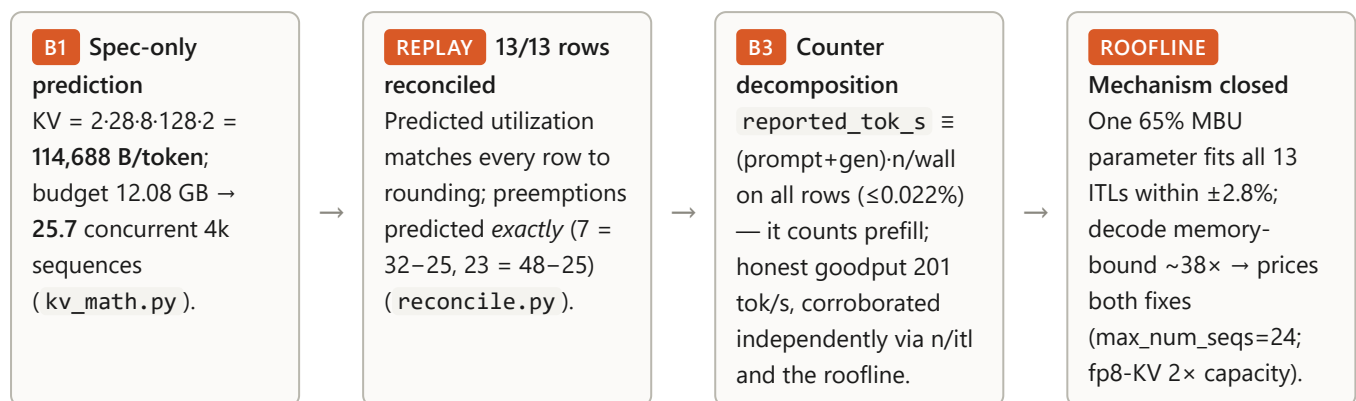
Core correction. A cross-language cost metric needs a denominator that holds *content* constant. Words, graphemes, code points and bytes all fail in different directions (per-word errs –15% for Hindi and +36% for Kannada on the same data); only tokens-per-parallel-sentence is the quantity cost scales with. The "property of the script" root-cause is falsified by a 6.4× premium spread with script and content held constant:

The 'Indic premium' is a tokenizer property, not a script property

Tokens needed for the same 1012 parallel sentences, relative to English — FLORES-200 devtest; dashed line = English (1.0)



4 Part B — capacity reconciliation as executable arithmetic



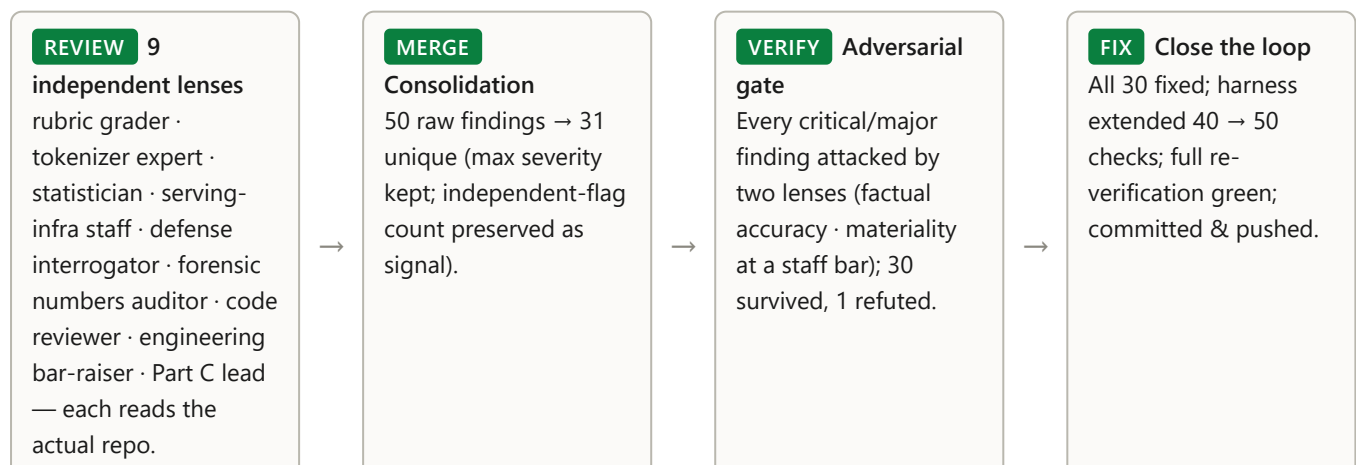
The reconciliation doubles as instrument calibration: it also settles that the stack's "24 GB" is decimal (a GiB reading predicts util 0.82 / 3 preemptions at the knee — rejected by the logged 0.93 / 7), and surfaces a spec inconsistency v0 never noticed (vocab = 128k, which is not gpt2 — REPORT_v1 §4 brackets the production premium and makes measuring the real tokenizer a recommendation).

5 Verification architecture

- **verify.py** — 50 assertions: baseline repro, per-bug deltas, corpus SHA-256 pins, the parity grid, prose-quoted derived ratios, romanization on two tokenizer generations, all 13 bench-row reconciliations, roofline fit. Exit code gates CI.
- **CI** (.github/workflows/verify.yml): runs the harness on every push.
- **Reproducibility**: fixed seeds, hash-pinned corpus, .gitattributes * -text (a CRLF checkout would silently shift token counts), shared _paths.py input resolver.
- **Honesty artifacts**: NOTEBOOK.md — chronological log with dead ends and self-caught errors kept; AI_USAGE.md — where AI helped, its seven concrete failure incidents, and the candidate's defense-prep checklist.
- **Figures** regenerate from data (figures/make_figures.py); annotations are computed, never typed.

6 Adversarial review architecture (round 3)

After two human rejections, the submission was audited the way it audits the intern — a deterministic multi-agent workflow:



59 agents, ~3.3M subagent tokens, 784 tool calls (scale logged in NOTEBOOK round 4). The findings and their fixes are themselves documented evidence (docs/INCIDENTS.pdf , NOTEBOOK rounds 3–4).

7 Repository map

```
submission/
  REPORT_v1.md          the product: corrected report for the leadership deck (figures/)
  verify.py             numbers-under-test harness – 50 assertions; defense entry point
  _paths.py             starter-kit resolver (env var / both zip nestings, fail-fast)
  NOTEBOOK.md · AI_USAGE.md · requirements.txt
  partA/ audit/ (repro, one-toggle experiments, findings, romanization) corpus/ (builder, card, 7
hash-pinned files)
    fertility_v1.py (5 tokenizers × 4 denominators, CIs, unk%) results/ memo.md
  partB/ kv_math.py · reconcile.py (13-row replay + roofline) · capacity.md
  partC/ memo.md        casual-tone decision memo (tie-proof gates, kill criteria)
docs/  ARCHITECTURE.pdf · INSTRUCTIONS.pdf · INCIDENTS.pdf (+ HTML sources, regeneration script)
.github/workflows/verify.yml · ARCHITECTURE.md · README.md · starter_kit/ (audited inputs)
```

Generated from the repository at commit history head · all numbers in this document are asserted by `python submission/verify.py` (ALL 50 NUMBERS VERIFIED).